

Summary

- 7+ years building ML infrastructure at scale, from model serving platforms to GPU training clusters managing 500+ GPUs.
- Experience spanning model hosting, LLM serving, training infrastructure, and developer tooling across Kubernetes and cloud environments.
- Open source contributor: Apache Airflow Taskflow API (shipped in 2.0), TFX, Triton Server.

Experience

- **Block Inc** *L6 Machine Learning Engineer, September 2024 - Present*
 - Designed LLM serving gateway with OpenAI-compatible API on the model hosting platform, enabling teams to host open-source models like GLM 4.6 via vLLM at scale, reducing integration time from weeks to days.
 - Built inference request archival and evaluation logging system, now logging 19B+ prediction records monthly at 7.5K rows/second across 145 production ML models for model debugging, statistical analysis, and quality assessment.
 - Originated Slurm-based ML training cluster as a hack week project, maintained it as production infrastructure for 2+ years enabling a cross-org research team's NeurIPS-accepted paper.
 - Built Kubernetes-based ML training platform managing 500+ GPUs across 15 accelerator types with 75+ monthly active users, replacing the legacy Slurm cluster and completing a full platform migration.
- **Block Inc** *L5 Machine Learning Engineer, March 2023 - September 2024*
 - Co-authored initial version for multi-model hosting endpoints reducing cost by 50%.
 - Co-author of Python SDK for automatic model deployment reducing deploy times to hours instead of days.
 - Developed initial support for hosting LLM within Block.
 - Presented in industry conference about scaling up model hosting at Block.
- **Twitter/X** *Senior Machine Learning Engineer, April 2022 - February 2023*
 - Co-authored and launched initial version of X Inc large scaling training cluster with Slurm and Ansible.
 - Technically led team of 8 engineers on ML Pipelines team, defining roadmap and execution cadence.
 - Worked with industry collaborators from Spotify and Google for alignment on reusable components for automatic model re-training pipelines.
- **Twitter** *Machine Learning Engineer II, October 2020 - April 2022*
 - Led adoption and migration to production model retraining scheduling framework on top of Kubeflow clusters.
 - Led development for comprehensive automatic CI/CD systems.
 - Collaborated with Ads team to launch Ads Click model in GCP with improved developer iteration, leading to faster model improvements.
- **Twitter** *Machine Learning Engineer, September 2019 - October 2020*
 - Co-authored internal hyperparameter optimization framework for using Bayesian optimization with Ax.
 - Supported ML research work for "Tuning Word2vec for Large Scale Recommendation Systems" (RecSys2020).
- **Project EPIC, University of Colorado Boulder** *Research Assistant, February 2017 - August 2019*
 - Analyzed 2000 tweets/minute of real time data by building Big Data pipeline using Spark, Kafka and Cassandra.
 - Deployed system prototype with Kubernetes using custom microservices written in Go and Python.
- **Twitter** *Software Engineering Intern, May 2018 - August 2018*
 - Developed CLI to deploy Apache Airflow instances internally.
 - Helped reduce ML model training and deploy time by 1/4 developing ML Workflows.
- **inLab FIB** *Web Development Intern, March 2015 - January 2017*
 - Built REST API for faculty data - Average usage of 25.000 daily accesses. (Django, Python)

Education

- **University of Colorado Boulder** Boulder, CO
M.S., Computer Science (GPA 3.97) *August 2017 - August 2019*
 - **Courses:** Analysis and Design of Algorithms, User Centered Design, Database Systems, Big Data Architecture, Machine Learning, Computational Genomics, Natural Language Processing.
- **Universitat Politècnica de Catalunya** Barcelona, Spain
B.S., Computer Science (8.8/10) *September 2013 - July 2017*
 - **Courses:** Data Mining, Data Compression, Programming Languages, Software architecture, Databases, Parallelism, Project Management, Web architecture, Android, NoSQL databases, Reflective programming.

Volunteering

- **TFX Addons SIG** *Core Contributor, May 2021 - March 2023*
 - Release manager from version 0.1.0 to date.
 - Contributor and maintainer for project CI/CD system, XGBoostEvaluator, FeastExampleGen and others.
 - Collaborator for ModelCardTooling migration mentoring Twitter team members to contribute to project.
- **AIP-31: Airflow functional DAG definition / "Taskflow API"** *Core Contributor, January 2020 - July 2020*
 - Proposed and implemented improvement for Airflow DAG user experience adding explicit XCom data.
 - Released in Airflow 2.0.0 and used in one of the main tutorials for the project.
- **HackCU, University of Colorado Boulder** *Tech director, February 2017 - April 2019*
 - Co-led Colorado's biggest student hackathon with 600 attendees and a 15 people core team.
- **HackUPC, Universitat Politècnica de Catalunya** *Co-Director, November 2015 - September 2018*
 - Co-led Spain's biggest student hackathon with 500 attendees and a 20 people core team.
 - Led web development team with 5 people for 2 years, developing the event website and registration platform.

Awards

- **Google OSS Peer Bonus for contributions to TFX and TFX Addons** (2022)
- **Denver Government award winner at ETHDenver.** (2018)
- **Grant winner at MD5 Physical Cyber Hackathon.** (2018)
- **Balsells Fellowship.** (2017-2019, two years of funding for graduate studies)
- **Balsells Mobility program.** (2017, five months of funding for research)
- **52North Innovation Incentive Award.** (2013)

Publications

- Jennings A., Gerard C., Kenneth M. A., Leysia P., and Rebecca M. **Incorporating Context and Location Into Social Media Analysis: A Scalable, Cloud-Based Approach for More Powerful Data Science.** *HICSS 52 (accepted).*
- Gerard C. **Social Media Analysis for Crisis Informatics in the Cloud** *MS Thesis*

Skills

- **Programming:** Python, Go, Terraform. **Infrastructure:** Kubernetes, EKS, Helm, ArgoCD, Slurm, vLLM, gRPC, AWS, GCP. **Data & ML:** Kubeflow Pipelines, Apache Airflow, Spark, Kafka.
- **Native:** Catalan, Spanish. **Advanced:** English.
- **Hobbies:** Skiing, Running, Photography, Roller hockey, Piano, Swim.